

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Even Semester)

Innovative Practices Description

UNIT II Oscillators

Degree, Semester & Branch: IV Semester B.E ECE B

Course Code & Title: EC8452 Electronic Circuits II

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: RC and LC Oscillators

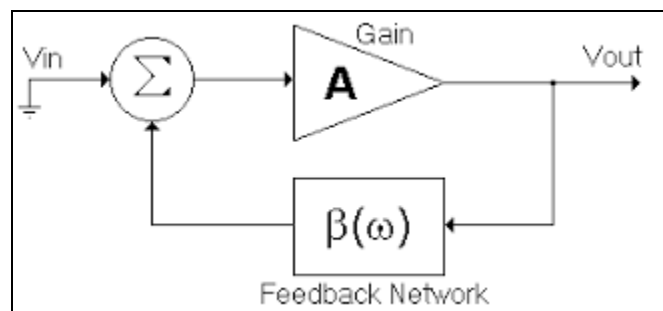
Name of the Innovative Practice: Theoretical concepts linked with Laboratory experiments

Description:

An oscillator is an electronic circuit for generating ac signal voltage with a dc supply as the only input requirement. The frequency of the generated signal is decided by the circuit elements. An oscillator requires an amplifier a frequency selective network and positive feedback from the output to the input. The Barkhausen criteria for sustained oscillator is $A\beta = 1$, where A is gain of the amplifier and β is the feedback factor.

If common emitter amplifier is used with resistive collector load, there is an 180° phase shift between input and output. The feedback network introduces an additional 180° phase shift at a particular frequency. There are different types of oscillator. RC oscillator is particularly useful as audio frequency generator where frequency determined network is resistor and capacitor. LC oscillator has LC tank circuit for frequency selection. LC oscillators are preferred for higher frequencies. Voltage divider bias is used for the amplifier in CE configuration. Amplifier section provides 180° phase shift. The tank circuit provides another 180° phase shift to satisfy Barkhausen criteria.

In Circuits and Simulation laboratory, students are exposed to design oscillator circuit for given frequency. Therefore, I have explained theoretical concepts of oscillator which linked with laboratory experiments and also explained that how to calculate the theoretical and practical value of frequency of generated signal using oscillator circuit in order to ensure the good design of circuit.



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	3	1	2	-	-	1	-	1	-	2

CO2: The student will be able to design different types of Oscillator at different frequency

References:

1. <http://www.sitttrkerala.ac.in/misc/LabManual/ECL.pdf>
2. Jacob Millman, “Microelectronics“, McGraw Hill, 2nd Edition, Reprinted, 2009.